

DATA SHEET

RDA5815S

**Fully Integrated, Direct Down-Conversion Satellite
Receiver for DVB-S,DVB-S2&ABS-S,MMDS**

Update History

Rev	Date	Author	History Description
1.0	2011-7-18	Wang hongxin	The primary datasheet
1.1	2011-7-29	Wang hongxin	Modify the application circuit to use the xtal pin as the reference clock for BB
1.2	2011-8-11	Wang hongxin	Modify the application circuit and pin description
1.3	2011-8-31	Wang hongxin	Modify the application circuit<add the input match circuit>
1.4	2011-9-5	Wang hongxin	Add the capacitance at the RFIP for mmDS application
1.5	2011-11-16	Yin xuesong	Add the junction temperature data and absolute maximum ratings data See table10 and table 11
1.6	2011-12-15	Wang hongxin	Add the max value of power consumption
1.7	2012-1-5	Wang hongxin	Modify the VCO&LNA description
1.8	2012-1-11	Wang hongxin	Modify the description of application circuit
1.9	2012-11-22	Wang hongxin	Revise some writing mistake

Features

- Single-chip RF-to-baseband Satellite receiver
 - CMOS Fully integrated RF front end
 - Low noise and wide dynamic range zero-IF receiver
 - Input frequency range: 250 to 2150 MHz
 - Input signal level: -100 to 5 dBm
 - More than 85dB gain control range
 - Fully integrated PLL (dividers, charge pump, phase & frequency detectors, loop filters, etc.)
 - Integrated RX VCO
 - Integrated baseband LPF with selectable cut-off frequency from 4MHz to 40MHz with 1MHz step
 - Integrated LNA with RF AGC
 - Integrated reference oscillator (27MHz is default)
 - I²C bus interface
 - Automatic gain control
 - 0.11μm RF CMOS technology
 - 3V to 3.6V operation
 - power consumption of less than 600mW
 - Lower profile packages:
 - RDA5815s 4×4 mm QFN24
- Pin assignment (24pin QFN 4×4 mm)**

Applications

- Set-top boxes
- Digital video recorders
- Digital television
- Satellite PC-TV
- SMATV trans-modulators (Satellite Master Antenna TV)

General Description

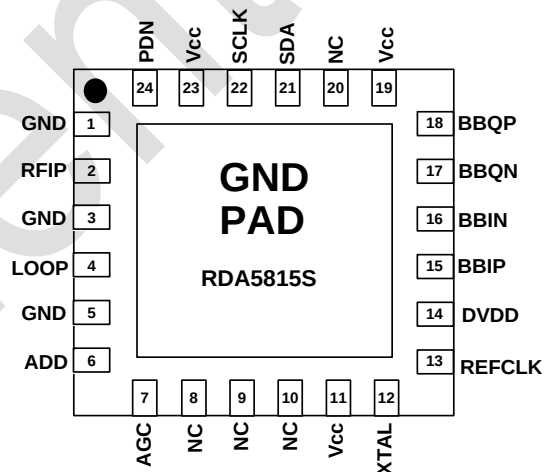
The RDA5815s is a fully integrated direct conversion

RF front end for DVB-S,DVB-S2&ABS-S,MMDS digital satellite

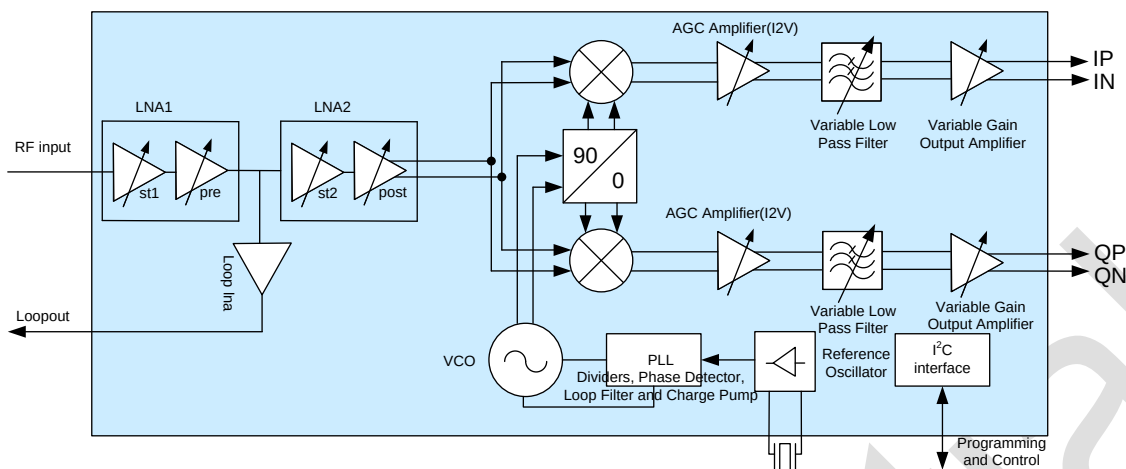
Reception standard CMOS process. The receiving frequency range is from 250MHz to 2150MHz, and the baseband filter's bandwidth can be selected from 4MHz to 40MHz with 1MHz step.

The rda5815s consists of a variable gain LNA, quadrature downconverter, variable IF gain amplifiers, variable low-pass filters, reference oscillator, VCOs, synthesizer and output baseband amplifier to drive external ADC.

Based on RDA's some innovative technique, the rda5815s offers excellent phase noise and very low implementation loss, required for advanced modulation systems such as 8PSK and DVB-S2. This tuner RF IC does not require a balun and its fully integrated design saves valuable board space and simplifies RF layout.



Functional Block Diagram



Functional Description

The loop Ina is used to generate the loopout signal. LNA1&LNA2 are on chip variable gain LNAs are used to provide wide tuner dynamic range. A direct conversion architecture is used to convert the RF signal to in-phase and quadrature baseband signals. The signals required for direct conversion are all generated within the chip by a fully integrated PLL and a quadrature LO generator. The frequency of the VCO is set by internal PLL circuits, which are programmable via a 2-wire (I²C) serial bus. The LO signals are mixed with the RF signal input and then filtered by low-pass filters to remove the upper image produced by the mixer. A variable gain amplifier is then used to adjust the baseband signal levels before processing by the channel selection filters to optimize noise performance and prevent distortion within the filters. The channel select filters are digitally programmable low pass filters with 1MHz step from 4MHz to 40MHz. The output amplifier buffered the signal from the filters to increase the driving capability to the next baseband ADC.

Some innovative technique is used to correct and alleviate DC offsets inherent in direct conversion mixers, the channel select filters and output buffers.

Loop LNA

The loopout Ina is used to generate the loopout signal.

LNA1&LNA2

LNA1 is the first Ina and receives a signal from the LNB through external matching circuit, the input resistance is matched to 75Ω coaxial cable. The LNA2 is the second Ina and is used to provide wide dynamic range, the better noise figure. The gain both of Ina1 and Ina2 can be programmed with the 2-wire (I²C) interface to insure wide dynamic range of the receiver.

AGC Amplifier

The AGC Amplifier receives the IQ signals from the quadrature mixer then amplifies them and sends to the base band low pass filter. The gain is also programmed and controlled by the 2-wire (I²C) interface.

Baseband LPF

The baseband low-pass filter is designed to have variable cut-off frequency. The bandwidth of the LPF is selectable from 4MHz to 40MHz with 1MHz step. The DC-offset calibration can be carried out controlled by the interface. An innovative technique ensure the filter's bandwidth variety within 5%.

Reference Oscillator

The oscillator is on-chip integrated and both the crystal and crystal oscillator are supported. The 27MHz is recommended. The centre oscillation frequency can be adjusted accurately with the XAFC pin (optional function).

VCO

The receiver integrates two VCOs which ensure covering the full receiving frequency from 250MHz to 2150MHz. The two VCOs can be switched between each other with relevant register and the VCO's band is programmed using 2-wire (I²C) interface. The first VCO oscillates from 3980MHz to 6290MHz, the second oscillates from 6000MHz to 8930MHz. After divided by four, eight or sixteen, lo signal from 250(4000/16)MHz to 2150(8600/4) can be obtained.

LO frequency	Used VCO	Divider	Note
$950(7600/8) \leq F \leq 1075(8600/8)$	VCO2	8	VCO1 oscillates from 3980MHz to 6290MHz. VCO2 oscillates from 6000MHz to 8930MHz
$1076(4304/4) \leq F \leq 1552(6208/4)$	VCO1	4	
$1553(6212/4) \leq F \leq 2150(8600/4)$	VCO2	4	

PLL

The fractional-N frequency synthesizer use 27MHz reference as default. The dividers, loop filters, phase&frequency detector, charge pump are all integrated on chip.

Baseband Output Amplifier

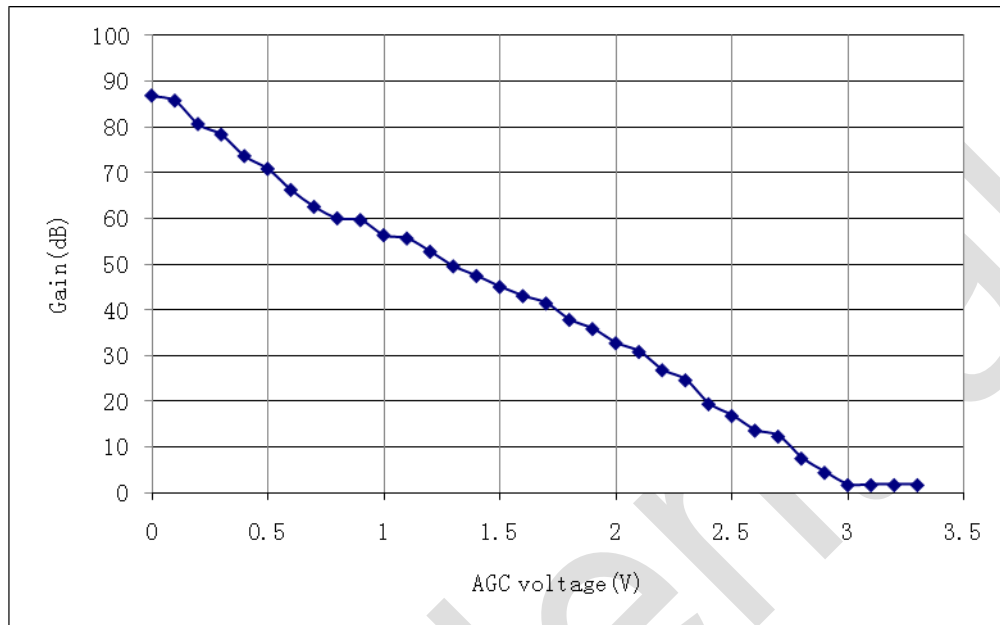
Class-AB architecture is selected for the output buffer, this makes it more flexible to the next Baseband solution's input load. The gain of this stage can also be programmed with interface to farther increase the receiver dynamic.

I²C interface

The interface is the control unit of all the analog blocks, it is provided for configuration and monitoring all internal registers, the I²C bus consists of two wires: serial clock line (SCL) and serial data line (SDA). The LNA's gain, AGC, baseband LPF's bandwidth, VCO's and crystal oscillator's current, VCO's oscillation frequency are all controlled by the 2-wire interface.

Direct AGC control

Through the AGC pin input, the receiver chain's gain can be directly controlled by the baseband solutions. The receiver gain changed inversely with the AGC control signal, this means that when AGC signal goes high, the receiver gain will drop to relevant level and vice versa.



Application circuit1 (for dvbs 950M~2150M)



- 1、The 5815s support the loopout, the setting is as follows:**

	rf_bypass_enable_b	0: power up 1: power down Default: 1'b0 The bypass buf functions at power-up of the chip by default. If this function is not required, set this bit to logic '1', which will save about 6mA current.
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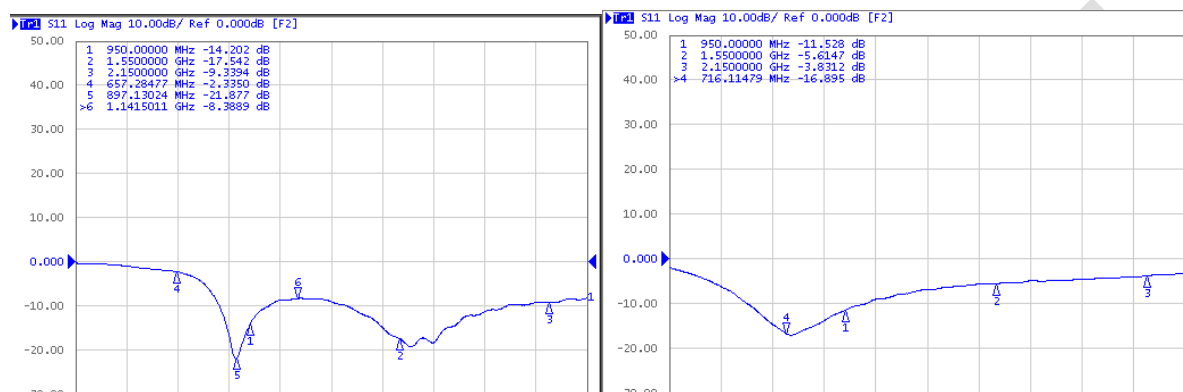
2、The input match circuit

This circuit is optional. If it is used or not, according to the performance requirement and test result.

There will have better S11, the noise figure of high frequency and reduce the low frequency block with input match circuit.

With input match circuit	Without input match circuit	Note
C1=2.7pF,C2=3.9pF L1=3.9nH,L2=5.6nH	C1=0ohm,C2=0ohm L1=NC,L2=NC	L1&L2 must be high frequency inductor and self-resonant frequency must greater than 3GHz

S11



S11 with input match circuit

S11 without input match circuit

Noise Figure

frequency	Noise Figure	
	with match	without match
1000	2.97	2.47
1050	2.96	2.66
1150	3.48	2.98
1250	3.32	2.92
1350	3.32	3.12
1450	3.17	3.37

frequency	Noise Figure	
	with match	without match
1550	3.36	3.76
1650	3.55	4.05
1750	3.74	4.24
1850	4.32	4.82
1950	3.72	5.42
2050	4.29	5.19
2150	4.73	5.93

Low frequency block

When frequency is less than 657MHz, the S11 is very worse. So the frequency which is less than 657MHz will be reject because that the input match circuit is a high-pass filter structure.

2、IQ filter circuit

This circuit is optional.

If this circuit is to be use or not according to the test result of different demodulator chip and pcb

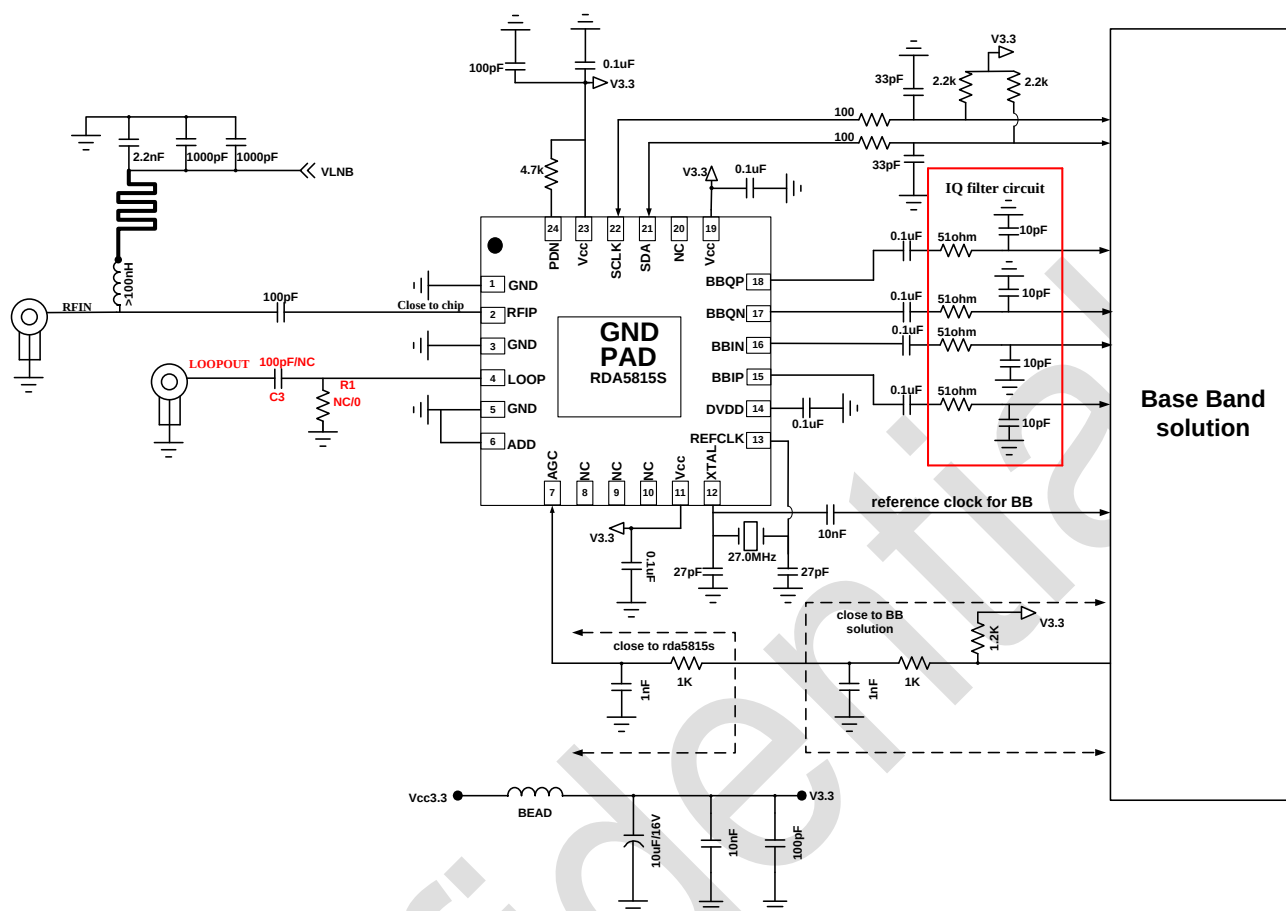
3、Crystal circuit

The value of load capacitance is determined by the crystal parameter.

The shared clk is optional according to the solution requirement.

So, the input match circuit, IQ filter circuit and the shared clk are optional. If these circuits are used or not, according to the solution requirement and test result.

Application circuit2 (for mmds 250M~2150M)



Note:

1、The 5815s support the loopout, the setting is as follows:

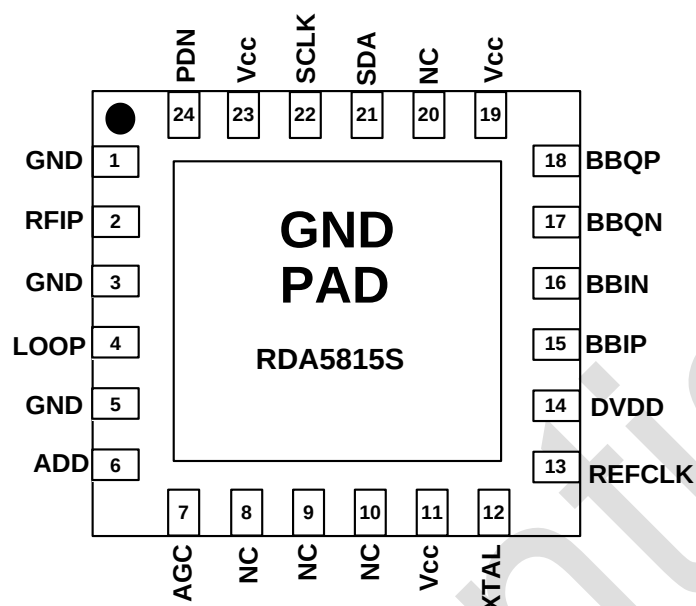
loopout	The third pin of 5815s<loop>	The value of register 04H's 5bit
on	R1=NC, C3=100pF	0
off	R1=0, C3=NC	1

5bit of register 04H	rf_bypass_enable_b	0: power up 1: power down Default: 1'b0 The bypass buf functions at power-up of the chip by default. If this function is not required, set this bit to logic '1', which will save about 6mA current.
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2、IQ filter circuit

If this circuit is to be use or not according to the test result of different demodulator chip and pcb.

Pin Description



Pin number	Name	I/O	DESCRIPTION
1	GND		
2	RFIP	I	RF Positive Input
3	GND.		
4	LOOP.	O	LOOPOUT
5	GND..		Not connected.
6	ADD	I	I2C bus address selection terminal. Allowing the use of more than one device per I2C bus system by the voltage on this pin. See Table 8 for programming details.
7	AGC	I	AGC control input from baseband solutions
8	NC		Not connected.
9	NC		Not connected.
10	NC		Not connected.
11	Vcc.	I	Supply voltage for 5815s. Connected to 3.3 V..
12	XTAL	I/O	Connect to Crystal (27MHz is recommended)
13	REFCLK	I/O	Connect to Crystal (if using external crystal oscillator, this pin connect to the oscillator's output)
14	DVDD	O	Output of the supply voltage for Digital, Connected to cap
15	BBIP	O	Positive Baseband I Channel Out
16	BBIN	O	Negative Baseband I Channel Out
17	BBQN	O	Negative Baseband Q Channel Out
18	BBQP	O	Positive Baseband Q Channel Out

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19	Vcc	I	Supply voltage for 5815s. Connected to 3.3 V.
20	NC		
21	SDA	I/O	Serial data input/output, connected to 3.3V with 10k Ω resistor
22	SCLK	I	Serial clock input
23	Vcc	I	Supply voltage for 5815s. Connected to 3.3 V.
24	PDN	I	Receiver power control If low, Receiver total power down, including the crystal oscillator. If High, the chip is enabled.

Electrical Specifications

Table 1 Recommended Operating Conditions

Parameter	Symbol	MIN	TYP	MAX	UNIT
Analog Supply Voltage	AVDD	3	3.3	+3.6	V
Ambient Temperature	T _A	-25	27	+85	°C

Table 2 DC Electrical Specification

Parameter	Symbol	MIN	TYP	MAX	UNIT
CMOS Low Level Input Voltage	V _{IL}	0		0.3*VDD	V
CMOS High Level Input Voltage	V _{IH}	0.7*VDD		VDD	V
CMOS Threshold Voltage	V _{TH}		0.5*VDD		V

Table 3 Power consumption specification

(VDD = 3 to 3.6 V, T_A = -25 to 85 °C, unless otherwise specified)

Symbol	Description	Condition	MIN	TYP	MAX	UNIT
ICC	Receiver on			170	175	mA
ICC(standby)	tuner Disabled	PDN=1,Enable=0,Rxon=0		6	7	mA

Table 4 System Characteristics

(VDD = 3 to 3.6 V, T_A = -25 to 85 °C, unless otherwise specified)

Parameter	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT
RF input frequency	f _{IN}		250		2150	MHz
Minimum RF input	RFIL	4.42MS/s, QPSK3/4, noise free		-97	-95	dBm
Maximum RF input	RFIH	4.42MS/s, QPSK3/4, noise free		5		dBm
Minimum RF input	RFIL	45MS/s, QPSK3/4, noise free		-87	-85	dBm
Maximum RF input	RFIH	45MS/s, QPSK3/4, noise free		5		dBm
Noise Figure	NF	Max Gain (Minimum AGC)		3.7	4.5	dB
Input referred third-order intercept	IIP3	Minimum Gain (Maximum AGC)		5	7	dBm
Input referred second-order intercept	IIP2	Minimum Gain (Maximum AGC)	41	47	55	dBm
Gain flatness over frequency	GF	Input freq=900 to 2250MHz		3	4	dB
IQ amplitude balance	IQAB			0.2	0.5	dB
IQ phase balance	IQPB			0.3	0.5	Deg
Minimum voltage	Gv(min)	AGCIN set to maximum		4		dB

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conversion gain		(3.3 V)				
Maximum voltage conversion gain	Gv(max)	AGCIN set to minimum (0.1V)		82		dB
Voltage conversion gain step	Gv(step)			0.2		dB
Matched input resistance	R _{IN}	After matching		75		Ω
Power up setting time	PUST			1		ms
Input reflection coefficient	S11	After matching		-8	-7	dB

Table 5 Frequency Synthesizer Characteristics

(VDD = 3 to 3.6 V, T_A = -25 to 85 °C, unless otherwise specified)

SYMBOL	DESCRIPTION	CONDITIONS	MIN	TYP	MAX	UNIT
t _{SW}	RX Switch on time			200		us
f _{RFO}	synthesizer frequency		900		2250	MHz
PN1	Phase noise	Δf=1kHz		-85	-75	dBc/Hz
PN2		Δf=10kHz	-100	-95	-90	dBc/Hz
PN3		Δf=100kHz	-110	-105	-100	dBc/Hz
PN4		Δf=1MHz	-138	-133	-130	dBc/Hz

Table 6 Baseband LPF and Output Amplifier Characteristics

Parameter	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT
3 dB baseband filter bandwidth	BBF(3dB)		4		40	MHz
Cutoff frequency accuracy at -3dB	FC				+/-5	%
LPF 2fc attenuation	LPF ATT1		25			dB
Flatness	FLTN			0.5	1	dBpp
Group delay	Td(g)			2		ps
Maximum differential output voltage		Clipping level		0.9*VDD		Vpp
Output common mode voltage			0.3*VDD	0.5*VDD	0.6*VDD	V
Differential Output offset voltage				30		mV
IQ output impedance			25	45	55	Ω
Output Amplifier minimum load impedance			1K			Ω
Output Amplifier maximum Load capacitance					10	pF

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Table7 XTAL Characteristics

(VDD = 3 to 3.6 V, T_A = -25 to 85 °C, unless otherwise specified)

Parameter	SYMBOL	MIN	TYP	MAX	UNIT
XTAL differential output voltage			680		mVpp
XTAL common mode voltage			0.55		V

Control Interface

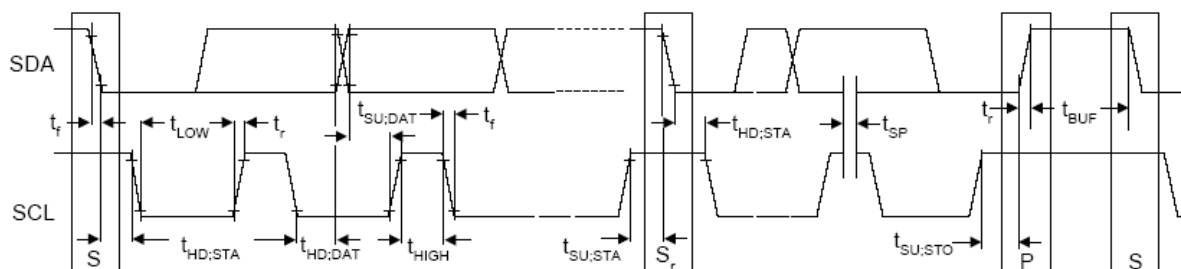
Table 8 I²C Address selection

MA1	MA0	ADD input voltage
0	0	0 V ~ 0.1 VDD
0	1	Open
1	0	0.4 VDD ~ 0.6 VDD
1	1	0.9 VDD ~ VDD

Table 9 I²C bus characteristics

(VDD = 3 to 3.6 V, T_A = -25 to 85 °C, unless otherwise specified)

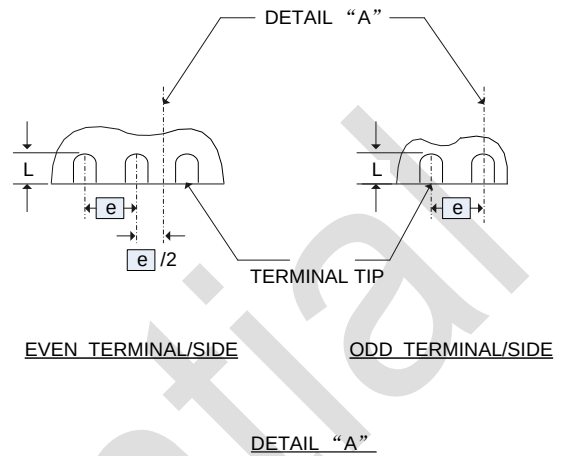
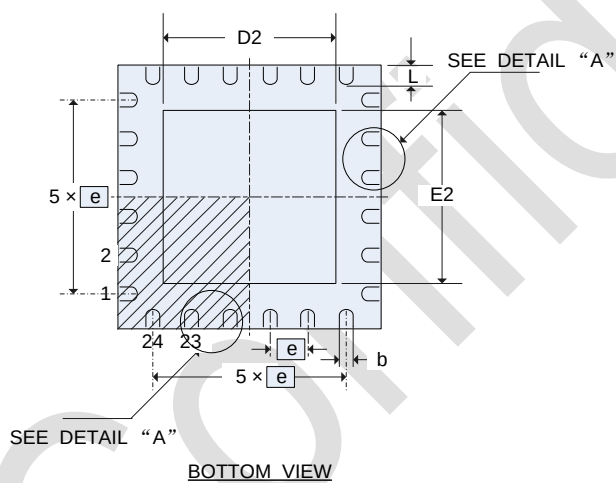
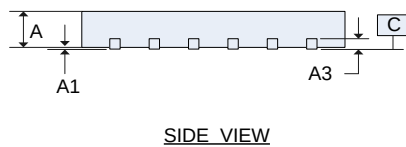
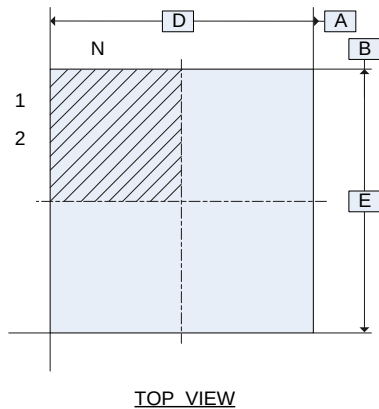
PARAMETER	SYMBOL	Test Condition	MIN	TYP	MAX	UNIT
SCL Clock Frequency	f _{SCL}		0		400	kHz
Bus Free Time between START and STOP Condition	t _{BUF}		1.3			μs
Hold Time (repeated) START Condition. (After this period, the first clock pulse is generated.)	T _{HD, STA}		0.6			μs
LOW Period of SCL Clock	t _{LOW}		1.3			μs
HIGH Period of SCL Clock	t _{HIGH}		0.6			μs
Data Setup Time	t _{SU,DAT}		100			ns
Data Hold Time	t _{HD,DAT}		0		0.9	μs
SCL and SDA Rise and Fall Time	t _r , t _f				300	ns
Setup Time for a Repeated START Condition	t _{SY,STA}		0.6			μs
Setup Time for STOP Condition	t _{SU,STO}		0.6			μs
Capacitive Load for each Bus Line	C _B				400	pF



I²C Timing Diagram

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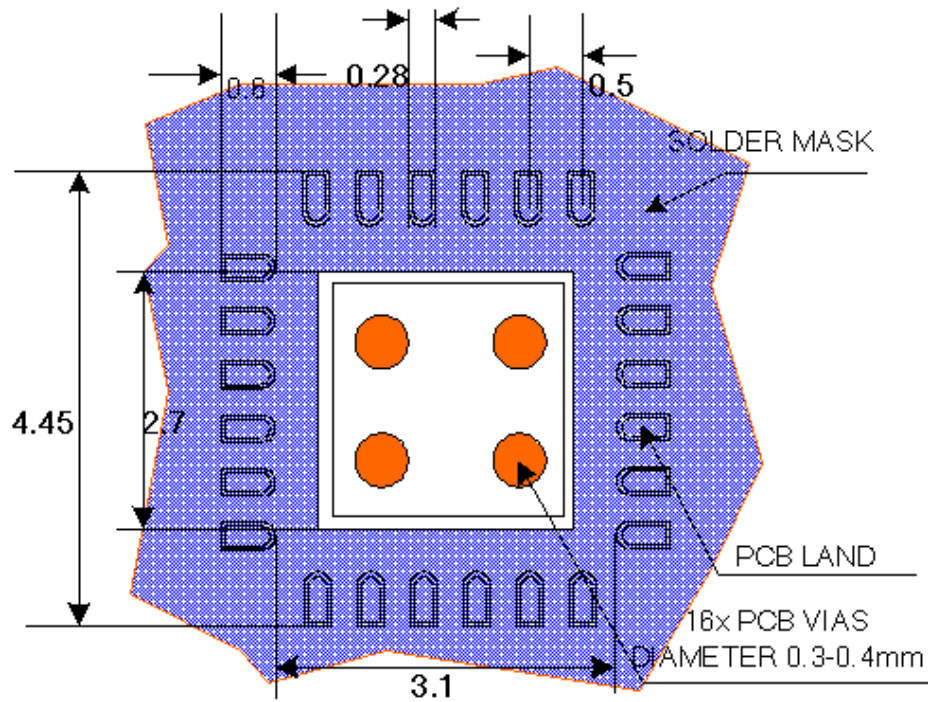
Package Outline



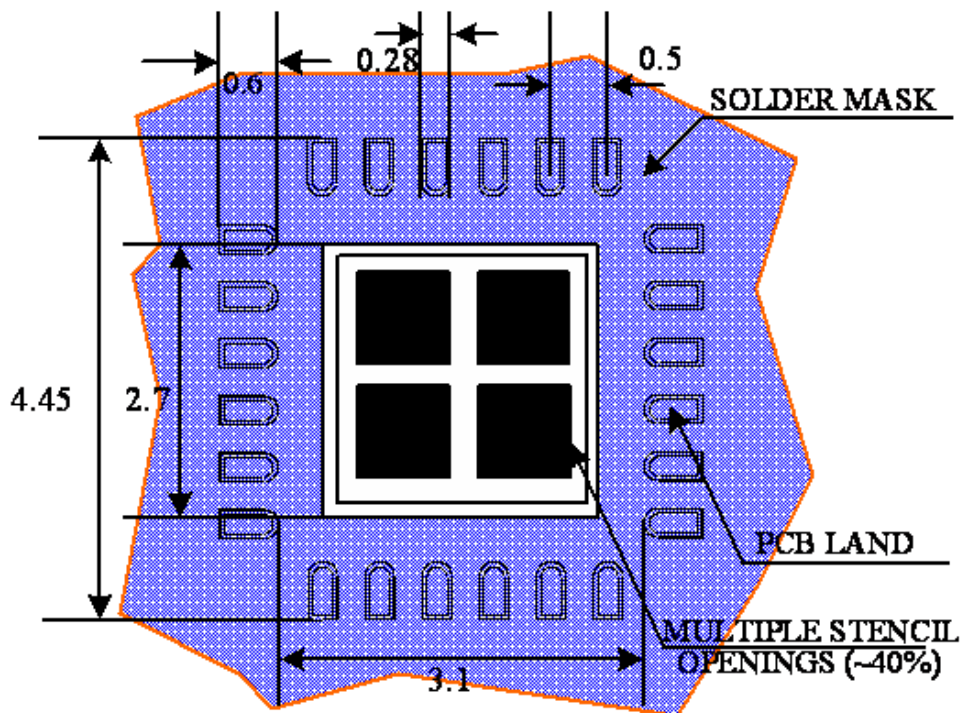
	MIN	NOM	MAX
D	4.00 BSC		
E	4.00 BSC		
D2	2.00	2.15	2.25
E2	2.00	2.15	2.25
e	0.50 BSC		
L	0.30	0.40	0.50
b	0.18	0.25	0.30
A	0.80	0.90	1.00
A1	0.00	0.02	0.05
A3	0.20 ref		

24-Pin 4x4 Quad Flat No-Lead (QFN)

PCB land pattern

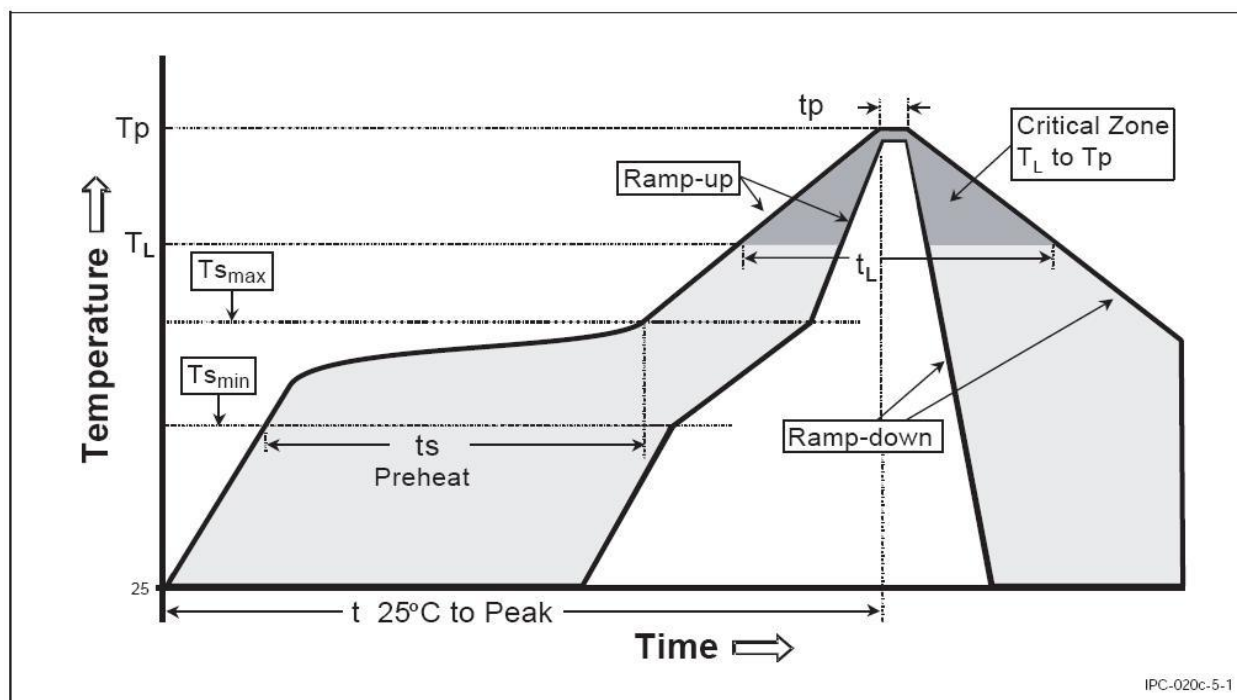


PCB Land Pattern for 24-Pin QFN



PCB Solder Paste Stencil Openings

Solder Mounting Condition



Classification Reflow Profile

Table-I Classification Reflow Profiles

Profile Feature	Sn-Pb Eutectic Assembly	Pb-Free Assembly
Average Ramp-Up Rate (T_{smax} to T_p)	3 °C/second max.	3 °C/second max.
Preheat		
-Temperature Min (T_{smin})	100 °C	150 °C
-Temperature Max (T_{smax})	100 °C	200 °C
-Time (t_{smin} to t_{smax})	60-120 seconds	60-180 seconds
Time maintained above:		
-Temperature (T_L)	183 °C	217°C
-Time (t_L)	60-150seconds	60-150 seconds
Peak /Classification Temperature(T_p)	See Table-II	See Table-III
Time within 5 °C of actual Peak Temperature (t_p)	10-30 seconds	20-40 seconds
Ramp-Down Rate	6 °C/second max.	6 °C/seconds max.

Time 25 °C to Peak Temperature	6 minutes max.	8 minutes max.
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Table – II SnPb Eutectic Process – Package Peak Reflow Temperatures

Package Thickness	Volume mm ³ <350	Volume mm ³ ≥350
<2.5mm	240 + 0/-5 °C	225 + 0/-5 °C
≥2.5mm	225 + 0/-5 °C	225 + 0/-5 °C

Table – III Pb-free Process – Package Classification Reflow Temperatures

Package Thickness	Volume mm ³ <350	Volume mm ³ 350-2000	Volume mm ³ >2000
<1.6mm	260 + 0 °C *	260 + 0 °C *	260 + 0 °C *
1.6mm – 2.5mm	260 + 0 °C *	250 + 0 °C *	245 + 0 °C *
≥2.5mm	250 + 0 °C *	245 + 0 °C *	245 + 0 °C *
*Tolerance : The device manufacturer/supplier shall assure process compatibility up to and including the stated classification temperature (this mean Peak reflow temperature + 0 °C. For example 260+ 0 °C) at the rated MSL Level.			

Note 1: All temperature refer topside of the package. Measured on the package body surface.

Note 2: The profiling tolerance is + 0 °C, - X °C (based on machine variation capability) whatever is required to control the profile process but at no time will it exceed - 5 °C. The producer assures process compatibility at the peak reflow profile temperatures defined in Table –III.

Note 3: Package volume excludes external terminals(balls, bumps, lands, leads) and/or non integral heat sinks.

Note 4: The maximum component temperature reached during reflow depends on package the thickness and volume. The use of convection reflow processes reduces the thermal gradients between packages. However, thermal gradients due to differences in thermal mass of SMD package may still exist.

Note 5: Components intended for use in a “lead-free” assembly process **shall** be evaluated using the “lead free” classification temperatures and profiles defined in Table-I II III whether or not lead free.

Table 10 Junction Temperature specification(VDD =3 to 3.6 V, T_A = -25 to 85 °C, unless otherwise specified)

Symbol	Description	Condition	MIN	TYP	MAX	UNIT
Θ _{JA}	Junction to Ambient Thermal Resistance			29.4		°C/W
T _j	Junction Temperature	Ta (Ambient Temperature) -25 °C	-10	-8.5	-7	°C
T _j	Junction Temperature	Ta (Ambient Temperature) 25 °C	40	41.5	43	°C
T _j	Junction Temperature	Ta (Ambient Temperature) 85 °C	100	101.5	103	°C

Note: T_j = T_a + Θ_{JA} * Power Consumption, MIN value comes from 3.0v VDD, TYP value comes from 3.3v VDD, MAX value comes from 3.6v VDD.

Table 11 Absolute Maximum ratings

Symbol	Description	MIN	MAX	UNIT
VDD	Power supply voltage	-0.3	3.6	V
T _{oper}	Operating ambient Temperature	-25	85	°C
T _j	Junction Temperature		125	°C

RoHS Compliant

The product does not contain lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyls (PBB) or polybrominated diphenyl ethers (PBDE), and are therefore considered RoHS compliant.

ESD Sensitivity

Integrated circuits are ESD sensitive and can be damaged by static electricity. Proper ESD techniques should be used when handling these devices.

Confidential

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